Scaling Retrieval Practice with LLM: Improving Multiple Choice Question (MCQ) Quality through Knowledge Graphs

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Abstract

Teaching introductory computer science has become increasingly difficult with the rise of AI code-completion tools. Frequent retrieval practice, especially through multiple-choice questions (MCQs), offers a promising way to maintain active learning, yet producing high-quality MCQs at scale remains a challenge for instructors. Large language models (LLMs) can automate MCQ generation, enabling scalable in-lecture retrieval practice. This poster presents two preliminary studies examining this potential. First, in higher-education programming courses, students who received LLM-generated MCQs scored significantly higher on follow-up quizzes than during periods without retrieval practice. However, raw LLM-generated MCQs often suffered from hallucinations, weak distractors, trivial content, and formatting issues. Second, we evaluated a knowledge graph (KG)–guided generation pipeline. By structuring key concepts and relations before prompting, the KG-based approach produced more relevant, integrative, and appropriately challenging MCQs. In a dataset of 400+ items, KG-generated MCQs outperformed text-only generation across 15 quality criteria. These preliminary studies open promising directions for future research, including adaptive generation of personalized MCQs tailored to student mastery levels.

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1 Introduction

Retrieval practice [7] as low-stakes formative assessment enhances long-term memory retention and overall learning [1, 5, 6]. In the context of introductory computer science courses, frequent retrieval practice during lectures can mitigate the challenges posed by the increasing prevalence of AI-based code auto-completion tools in programming exercises. Multiple choice questions (MCQs) are an easy way to conduct retrieval practice with quantitative results for analysis. Yet, the manual creation of effective multiple-choice questions (MCQs) is time-intensive, limiting their frequent use. Large language models (LLMs) offer a solution by generating MCQs directly from lecture materials (see Figure 1).

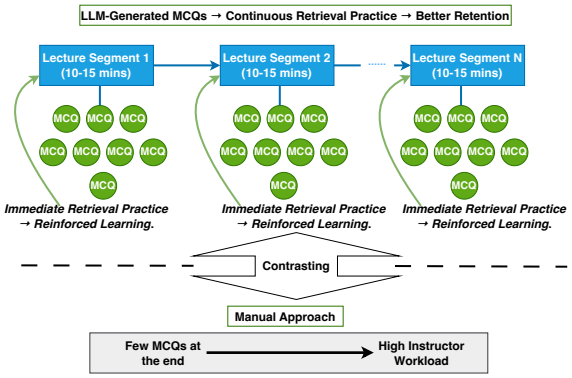

Figure 1: LLM-generated MCQs allow frequent, low-stakes retrieval practice throughout a lecture, compared to limited manually authored questions

Our classroom study demonstrates that such LLM-generated MCQs significantly improve student performance. However, raw generation presents critical quality issues: instructors often need to filter two-thirds of MCQs to identify usable MCQs. This bottleneck undermines scalability and reduces adoption.

We therefore ask: How can we retain the learning benefits of LLM-generated MCQs while improving their reliability and reducing instructor workload?

2 Methods

In two introductory programming courses, we conducted a preliminary study on the effectiveness of LLM-generated MCQs. We compared student performance during periods when retrieval practice with LLM-generated MCQs was provided against periods without such practice. Follow-up quiz scores were analyzed using Mann–Whitney U tests.

2.1 Quality Issues of Raw MCQs

We had to manually verify all LLM-generated MCQs before presenting them to students. Our analysis revealed several quality issues, including:

- Factual mistakes and hallucinations
- Duplicated or trivial questions
- Ambiguous options or questions

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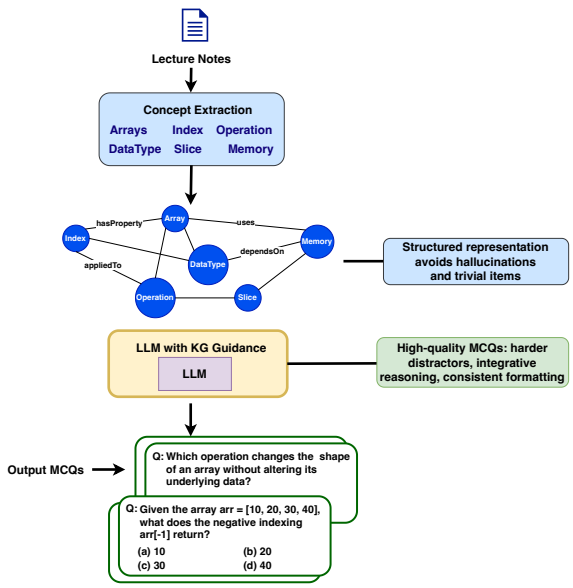

Figure 2: Knowledge Graph-Guided Pipeline for MCQ Generation

- Easy distractors and inconsistent formats

2.2 Knowledge Graph-Guided MCQ Generation Pipeline

A great deal of effort has been put into developing LLM-based systems for MCQ generation [3]. These models typically take a passage or concept description as input and generate question stems, correct answers, and distractors through prompt engineering or fine-tuning. Despite these advances, existing methods rarely leverage knowledge graphs for quality control, with only a few attempts to align MCQs with learning objectives [4].

To address these quality issues, we developed a pipeline that represented the lecture content as a knowledge graph (KG) and generated MCQs from it (see Figure 2). The process begins with a novel framework for constructing and refining knowledge graphs from lecture notes, grounded in the principles of Rate-Distortion theory and Optimal Transport [2]. By formalizing KG extraction as a compression problem, the framework generates a balanced knowledge graph, which is compact yet faithful to the lecture content. With this pedagogically grounded structure in place, the system prompts an LLM using targeted KG context to generate high-quality MCQs that align with the course content and maintain conceptual coherence.

3 Preliminary Findings

Effectiveness of LLM-Generated MCQs: Students using LLM-generated MCQs achieved a 16-percentage-point higher average on quizzes (89% vs. 73%), a statistically significant ($p < 0.0001$, $r = 0.586$) improvement.

MCQ Quality Evaluation: We evaluated approximately 200 KG-based MCQs against 200 raw MCQs derived directly from lecture notes, assessing them across 15 criteria using a 1–5 Likert scale. The KG-based MCQs consistently outperformed the raw MCQs across all lectures, yielding an average quality score improvement of approximately 15%. The largest gains occurred in `no_answer_hints`, `hard_to_guess`, and `no_duplicates`.

4 Discussion and Applications

Our preliminary findings illustrate both the promise and pitfalls of LLM-generated MCQs. Quality control is essential to reduce instructor workload and align with the learning objectives. A knowledge graph-guided MCQ generation pipeline is a promising solution. By grounding questions in structured representations, the pipeline can produce higher-quality items with stronger conceptual integration. This approach also opens the door to personalized MCQs that adapt to individual students' mastery levels.

The main challenge lies in developing an advanced KG-guided MCQ generation method that fully leverages the fine-grained knowledge in a task-oriented KG while supporting downstream assessment functions such as generating MCQs with calibrated difficulty and varied cognitive levels. Such a system must also track both aggregated and individual student mastery over time, allowing the MCQ generation process to adapt dynamically to each learner's evolving knowledge state.

5 Conclusion and Future Work

LLM-generated MCQs represent a powerful tool for scalable retrieval practice. Our preliminary experience confirms their effectiveness in improving student learning, while our KG-guided pipeline demonstrates that knowledge graphs can systematically improve their quality. Future work will focus on:

- Applying methods in psychometrics to item calibration.
- Maintaining instructor agency in MCQ generation and deployment.
- Generating individualized MCQs based on students' mastery levels.
- Conducting long-term retention studies.

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